A Mathematical Compendium of the Chiron / Primus Portfolio

Every formal object, with honest epistemic status

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Abstract

This document collects, in one place and in formal notation, the mathematics implemented or proposed across the portfolio: the exact invariant-recovery engine (Chiron), the Semantic Invariant Calculus (semic), Holographic Continuity Theory and its Projection Calculus (HCT), the stochastic-metriplectic physical substrate (UMA / RSLS), the governance and decision calculus (JDICert / SoCPM), the ambiguity calculus (Infectatrum), the derived measures used by the bridge modules (aesthetics, value, language), the combinatorics of the Caramuel atlas (Primus), and the variational Projection–Innovation Hierarchy (PIH). Each section is tagged with an explicit epistemic status. The unifying object is a single operation — recover the minimal generator of a surface under a description-length criterion, verify it by exact held-out prediction, and *refuse* (INCOMPRESSIBLE) when no generator in the hypothesis class is confirmed — applied across integers, ciphers, code, language, dynamics, and law.

Epistemic-status legend.

- *Standard result.* A known theorem or definition from the literature, used as-is.
- *Implemented & tested.* Computed exactly in code and covered by passing self-tests/benchmarks.
- *Proof-of-concept.* Implemented and internally self-tested; not validated on external/real-world data.
- *Self-developed theory.* Author-developed; constructive but not peer-reviewed.

Contents

1 The Invariant Engine (Chiron)

The engine treats every observable as a *codified surface* s (an integer sequence, a string, a ciphertext, a graph, source code) and seeks the shortest program that regenerates it.

1.1 Minimum Description Length

Definition 1.1 (Two-part code)**.** For a surface s and a candidate generator (model) M, the description length is

$$L(s, M) \;=\; L(M) \;+\; L(s \mid M),$$

where $L(M)$ is the bit-cost of specifying M (its class and exact rational parameters) and $L(s \mid M)$ is the bit-cost of the residual left after M regenerates s.

Definition 1.2 (Collapse)**.** Let $\mathcal{H} = \bigcup_k \mathcal{H}_k$ be the union of hypothesis classes (Section 1.4). The recovered invariant is

$$M^\star \;=\; \underset{M \in \mathcal{H}}{\arg\min} \;\big[\, L(M) + L(s \mid M) \,\big],$$

computed in exact rational arithmetic (M's parameters lie in $\mathbb{Q}$ via `fractions.Fraction`). Ties are broken toward the canonically simpler class, and the margin to the runner-up is reported.

This is a tractable, exact restriction of the Kolmogorov / Solomonoff ideal (shortest generating program, uncomputable in general) to decidable classes, scored by MDL (Rissanen).

Status: *Implemented & tested. Standard MDL/Kolmogorov lineage.*

1.2 Held-out verification and abstention

Definition 1.3 (Held-out gate)**.** Split s into a training prefix $s_{1:t}$ and a withheld tail $s_{t+1:n}$. $M^\star$ recovered from $s_{1:t}$ is *verified* iff it predicts the tail at exact equality,

$$M^\star(j) = s_j \quad \text{for all } j > t \quad \text{(equality, not tolerance).}$$

If no candidate verifies, the engine returns a classified residual and the verdict INCOMPRESSIBLE.

This zero-tolerance gate is the load-bearing property; it yields the measured zero-false-positive behaviour (no spurious fit on a prefix survives the withheld continuation).

Status: *Implemented & tested.* $\approx 5{,}000$*-case fuzz with* 0 *false verifications.*

1.3 Compression ratio

Definition 1.4. With $b_{\text{raw}}(s)$ the verbatim bit-size and $L(s, M^\star)$ the two-part code length,

$$\rho(s) \;=\; \frac{b_{\text{raw}}(s)}{L(s, M^\star)} \;\geq\; 1, \qquad \text{(saturation } \mathcal{E} = 1 - \rho^{-1} \in [0,1]).$$

$\rho \approx 1$ signals no exploitable structure; large ρ signals a tiny law generating a large surface.

Status: *Implemented & tested.*

1.4 The hypothesis classes

Each class is a closed-form generator fitted exactly over $\mathbb{Q}$.

$$\begin{aligned}
\text{Constant:}\quad & a_n = c. && (1)\\
\text{Arithmetic:}\quad & a_n = a_0 + d\,n. && (2)\\
\text{Geometric:}\quad & a_n = a_0\,r^{\,n}. && (3)\\
\text{Polynomial } (\deg k):\quad & a_n = \sum_{i=0}^{k} \binom{n}{i}\,\Delta^i a_0, \quad \Delta^i a_0 \text{ the forward differences.} && (4)\\
\text{Linear recurrence (order } m):\quad & a_n = \sum_{i=1}^{m} c_i\,a_{n-i}, \quad \chi(x) = x^m - \textstyle\sum_i c_i x^{m-i}, && (5)\\
& a_n = \sum_{j} p_j(n)\,\rho_j^n \ \ (\rho_j \text{ roots of } \chi). &&\\
\text{Periodic (period } p):\quad & a_n = a_{n \bmod p}. && (6)\\
\text{Multiplicative:}\quad & a_n = a_{n-1}\,f(n) \quad (\text{e.g. } a_n = n!). && (7)\\
\text{Alternating:}\quad & a_n = (-1)^n b_n. && (8)\\
\text{Holonomic / } P\text{-recursive (order } r):\quad & \sum_{i=0}^{r} p_i(n)\,a_{n+i} = 0, \quad p_i \in \mathbb{Q}[n]. && (9)
\end{aligned}$$

The holonomic class captures e.g. the Catalan numbers, $C_{n+1} = \dfrac{2(2n+1)}{n+2}\,C_n$, and the central binomial coefficients. An *interleaved composite* of k lanes assigns a_n to the generator of lane $n \bmod k$. Surfaces compressed by none of these are returned as a classified residual.

Status: *Implemented & tested.* 22/22 *in-scope OEIS-core sequences recovered,* 7 *correctly refused.*

1.5 Decoding as inverse recovery

A cipher is recovered by inverting the transform that produced it: given ciphertext c, find the transform T (and key) with $T^{-1}(c)$ a high-likelihood plaintext under the engine's language model; a known plaintext/ciphertext pair recovers the key directly. Classical schemes only (Caesar/affine/substitution/transp
n/Morse), not modern cryptography.

Status: *Implemented & tested.* 42/44 *classical instances solved ciphertext-only.*

2 The Semantic Invariant Calculus (semic)

The Chiron spine, lifted from integer sequences to meaning, in exact arithmetic.

2.1 Roles, edges, skeletons

Fix finite role vocabularies: operators $\{\textsc{transfer}, \textsc{construct}\}$, objects $\{\textsc{res}, \textsc{capc}\}$, horizons $\{\textsc{bounded}, \textsc{unbounded}\}$. A *skeleton edge* is a morphism operator(object) $\to$ horizon. A deterministic map skel : text $\to 2^{\text{Edge}}$ (a lexicon + word-order rule) sends a surface to its edge set; e.g. the "teach a man to fish" family gives $\{\textsc{transfer} \to \textsc{bounded},\ \textsc{construct} \to \textsc{unbounded}\}$.

Status: *Self-developed theory; the map is implemented & tested (reproducible, no model).*

2.2 MDL invariant, refusal, and the separable solver

Definition 2.1. For a family $F = \{O_1, \dots, O_n\}$ of surface edge-sets, with $L(S) = |S|$ and structural residual $\mathrm{Res}(O, S) = |O \triangle S|$,

$$I(F) = \arg\min_{S \subseteq U} \Big[\alpha\, L(S) + \sum_i \mathrm{Res}(O_i, S)\Big], \qquad \alpha \in \mathbb{Q}_{>0},$$

where $U = \bigcup_i O_i$ is the (finite) edge universe.

Definition 2.2 (Held-out refusal)**.** $I(F)$ is *verified* iff for every i the generator recovered from $F \setminus \{O_i\}$ predicts O_i with $\mathrm{Res} = 0$; otherwise the verdict is INCOMPRESSIBLE.

Theorem 2.1 (The collapse is separable: exact arg min in $O(|U|\,n)$)**.** *The energy $E(S) = \alpha|S| + \sum_i |O_i \triangle S|$ decomposes over edges. Writing $c_e = \big|\{\, i : e \in O_i \,\}\big|$ for $e \in U$,*

$$E(S) = \sum_{e \in U} g_e\big([e \in S]\big), \qquad g_e(1) = \alpha + (n - c_e), \quad g_e(0) = c_e,$$

a sum of independent *per-edge terms. Hence the exact minimizer is closed-form,*

$$S^\star = \Big\{\, e \in U : \ c_e > \tfrac{n+\alpha}{2} \,\Big\},$$

computable in $O(|U|\,n)$ — the identical generator that the full $O(2^{|U|})$ subset-lattice search returns.

Proof. $|O_i \triangle S| = \sum_{e \in U} \big([e \in O_i] \oplus [e \in S]\big)$ and $|S| = \sum_e [e \in S]$, so $E(S) = \sum_e g_e([e \in S])$ with the stated g_e. A sum of terms each depending on a single independent binary choice is minimized term-by-term: include e iff $g_e(1) < g_e(0)$, i.e. $\alpha + (n - c_e) < c_e \iff c_e > (n + \alpha)/2$. □

The exhaustive subset-lattice search is retained as a verification *oracle* (and the Gibbs and free-energy integrals of Theorem 2.2 still range over the whole lattice); the closed-form solver agrees with it on every gate family and on 3,003 random families across four values of α. The often-cited $O(2^N)$ cost is real but avoidable — this objective was never exponential.

Status: *Implemented & tested (*56/56 *gates).*

2.3 Typed hypothesis classes and constraint discovery

The semantic search is organized into bounded, typed classes $\mathcal{H}_1, \dots, \mathcal{H}_5$ — graph generators, logical rules, causal templates, organizational schemas, and semantic grammars — each a polynomially-bounded sub-lattice of 2^U. Collapse ranks the best generator *within* each class and then selects *across* classes, breaking ties toward the more constrained (more falsifiable) class, so MDL operates over a *defined* search space rather than an open universe. *Constraint discovery* then induces the functional predicates the recovered generator obeys (e.g. TRANSFER → BOUNDED), ranked by the fraction of the latent operator × horizon vocabulary each one removes.

Status: *Hypothesis classes implemented & tested; constraint discovery proof-of-concept.*

2.4 The bridge theorem (deterministic $= T \to 0$ stochastic)

Theorem 2.2. *Let $E(S) = \alpha L(S) + \sum_i \mathrm{Res}(O_i, S)$ on the finite generator space, and let $S^\star = \arg\min_S E(S)$ be unique. The Gibbs measure $p_T(S) = e^{-E(S)/T}/Z(T)$ satisfies $p_T(S^\star) \to 1$ as $T \to 0^+$ and $p_T \to$ uniform as $T \to \infty$.*

Proof. Write $\Delta(S) = E(S) - E(S^\star) \geq 0$ with $\Delta(S) > 0$ for $S \neq S^\star$. Then $p_T(S^\star) = \left(\sum_S e^{-\Delta(S)/T}\right)^{-1}$. As $T \to 0^+$ each $e^{-\Delta(S)/T} \to 0$ for $S \neq S^\star$, so the sum $\to 1$ and $p_T(S^\star) \to 1$. As $T \to \infty$ every $e^{-\Delta(S)/T} \to 1$, giving the uniform measure. □

The deterministic MDL collapse is thus the $T = 0$ limit of the stochastic (energy-based) collapse: one operator, two temperatures. With free energy $F(T) = \langle E \rangle_T - T\,H(p_T)$ one has $F(T) \leq \min_S E(S)$ with equality at $T = 0$, so the MDL generator is the zero-temperature free-energy minimum.

Status: *Self-developed theory; the limit is verified numerically over the exact space.*

2.5 Curvature of the collapse landscape

The generator space is the subset hypercube Q_n with Hamming distance $d(S, S') = |S \triangle S'|$. For a Markov kernel $\{\mu_S\}$ (a descent-biased lazy walk) the Ollivier–Ricci curvature on an edge (x, y) is

$$\kappa(x, y) \;=\; 1 - \frac{W_1(\mu_x, \mu_y)}{d(x, y)},$$

with W_1 the 1-Wasserstein distance, computed *exactly* via integer min-cost flow over rational measures (no floats). On the Congress graph of readings, a proverb is a clique with $\kappa > 0$ and an entendre is a barbell whose bridge edge carries genuine negative curvature:

$$\kappa(\text{entendre bridge}) = -\tfrac{1}{3}, \quad \kappa(\text{reading interior}) = +\tfrac{3}{4}, \quad \kappa(\text{proverb clique}) = +\tfrac{2}{3}.$$

Status: *Self-developed theory; curvatures computed exactly (rational).*

2.6 Functoriality, performatives, action

Domain invariance (Class VI). For a relabelling analogy φ between domains, collapse$(\varphi(s)) =$ collapse$(s) = P$: collapse is a natural transformation from the surface-realization functor to the constant functor at P (the naturality square commutes).

Performatives (Class VIII). A speech act is modelled as a state map on a world $W = (\text{obligations}, \text{roles}, \text{facts}, \text{aut}$ with felicity preconditions (Austin); promise, resign, declare change W when felicitous and *misfire* (no-op) otherwise.

Action principle. A trajectory through generator space has discrete action $\mathcal{S}(\gamma) = \sum_t \left[d(\gamma_t, \gamma_{t+1}) + \lambda E(\gamma_t)\right]$; the energy-monotone descent path is the action minimizer (verified by enumerating geodesics).

Twin proof. Two lexically disjoint surfaces collapse to one generator P over a combinatorial space of $6 \cdot 8 \cdot 4 \cdot 5 \cdot 5 \cdot 4 = 19{,}200$ realizations — the semantic analogue of the Caramuel anchor.

Status: *Self-developed theory; all four implemented & tested.*

3 Holographic Continuity Theory and the Projection Calculus (HCT)

HCT formalizes identity preserved under transformation. The reference implementation is `cert_engine.py` (280 tests).

3.1 Primitive objects

Definition 3.1. A *structure* is $S = (D, \mathcal{O}_D, \sim_D)$: descriptive content, an admissible operator family, and a recognition equivalence. Operators $\mathcal{O} : \mathcal{S} \to \mathcal{S}$ form a non-commutative semigroup under composition with identity $\mathbf{1}$ and partial inverses. The *orbit* is $\mathrm{Orb}_{\mathcal{F}}(S) = \{\mathcal{O}_n \circ \cdots \circ \mathcal{O}_1(S)\}$; *provenance* $\mathfrak{p}(S')$ is the operator word in the free monoid $\mathcal{F}^*$ producing S'. A *projection* $\pi : \mathcal{S} \to D_i$ is many-to-one with partial inverse (*recovery*) ρ. S is *identity-bearing* if $\rho(\pi(\mathcal{O}(S))) \sim \mathcal{O}(S)$ for all admissible $\mathcal{O}$.

Definition 3.2 (Holographic redundancy)**.** S has redundancy with threshold $\theta \in (0, 1)$ if any subset Σ of locals with $|\Sigma| \geq \theta|S|$ admits $\rho_\Sigma(\Sigma) \sim S$.

Definition 3.3 (Superposition state)**.** $\rho = \sum_k p_k \, |I_k^\star\rangle\langle I_k^\star|$, a density matrix over plausible invariant orbits.

Status: *Self-developed theory; primitives implemented & tested.*

3.2 Axioms (A1–A12, condensed)

Axiom. **A1** a non-trivial invariant orbit exists; **A2** the orbit class is preserved under admissible action; **A3** provenance is monotone non-decreasing, $\mathfrak{p}(S') = \mathfrak{p}(S) \cdot \mathcal{O}$; **A4** operators are prior to objects; **A5** significance is a function of relational-geometry curvature; **A6** every sufficient local participates in generating the global; **A7** agency exists at each scale; **A8** operators recur across scales via composition-preserving morphisms; **A9** superposition uses an augmented state $(\rho, \mathfrak{p})$ separating Markovian density from a deterministic provenance ledger; **A10** orbits generate novelty, not only preserve it; **A11** operator selection is stochastic, $H(\mathcal{O}) > 0$ (generic regime); **A12** significance is defined on the metric completion of the relational space.

Status: *Self-developed theory.*

3.3 Theorems (T1–T8)

Theorem 3.1 (T1 existence/uniqueness up to isomorphism)**.** *Under A1, an invariant orbit exists, and orbits under isomorphic operator algebras are isomorphic as orbit structures (the construction $\Phi(\mathcal{O}_n \cdots \mathcal{O}_1(S_1)) = \varphi(\mathcal{O}_n) \cdots \varphi(\mathcal{O}_1)(S_2)$ is a well-defined bijection respecting the orbit relation).*

Theorem 3.2 (T2 identity persistence)**.** *If S is identity-bearing and $\mathcal{O}$ admissible, $\mathcal{O}(S)$ is identity-bearing with triple $(\pi' = \pi \circ \mathcal{O}^{-1},\ \rho' = \mathcal{O} \circ \rho,\ \sim)$.*

Theorem 3.3 (T3 provenance conservation, generic regime)**.** *If $H(\mathcal{O}) > 0$ and provenance is path-injective, no admissible transformation strictly decreases $H(\mathfrak{p})$; identity-bearing structures cannot lose provenance information.*

Theorem 3.4 (T4 non-commutativity)**.** *On coupled Mystery Vectors at least one pair of epistemic operators has $[\mathcal{O}_i, \mathcal{O}_j] = \mathcal{O}_i\mathcal{O}_j - \mathcal{O}_j\mathcal{O}_i \not\equiv 0$ (measure-then-reframe $\neq$ reframe-then-measure when ignorance and paradox are correlated).*

Theorem 3.5 (T5 significance as scalar curvature)**.** *On the metric completion, any significance function is $\sigma(S) = f(\kappa(S))$ with f monotone and κ the scalar Ollivier–Ricci curvature; the reference implementation uses $\sigma = \tanh(\alpha\,\kappa)$.*

Theorem 3.6 (T6 multi-scale recurrence)**.** *There is a faithful, composition- and identity-preserving functor $\mathfrak{S} : \mathcal{F}_{s_1} \to \mathcal{F}_{s_2}$ between operator families at different scales.*

Theorem 3.7 (T7 holographic wholeness)**.** *For an admissible subset family, any σ with $|\sigma| \geq \theta|S|$ recovers S to recognition equivalence, and the underlying code rate is $\geq 1-\theta$ (Singleton bound).*

Theorem 3.8 (T8 Lindblad necessity)**.** *The marginal density of a superposition state, required to stay Hermitian, positive semidefinite, and unit-trace at all times, evolves by a GKSL master equation*

$$\frac{d\rho}{dt} = -i[H,\rho] + \sum_k \left(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\} \right),$$

not by classical mixture dynamics (which would violate complete positivity).

Status: *Self-developed theory; T1–T8 have constructive (informal) proofs, encoded as tests.*

3.4 The six epistemic operators and the action principle

On the Mystery Vector $\xi = (\xi_i, \xi_p, \xi_t, \xi_e, \xi_s, \xi_\infty) \in \mathbb{R}^6$ (ignorance, paradox, transcendence, emergence, subjectivity, infinity), the operators $\{\mathcal{M}, \mathcal{R}, \mathcal{I}, \mathcal{O}, \mathcal{D}, \mathcal{E}\}$ act as 6×6 contraction matrices each preferentially shrinking one coordinate. The HCT dynamics extremize

$$\mathcal{A}[S, \bar{\mathfrak{p}}] = \int \left(T(\dot{S}) - V(S) - \lambda H(\bar{\mathfrak{p}}) - \mu \, \|\dot{\bar{\mathfrak{p}}}\|^2 \right) dt,$$

with $\bar{\mathfrak{p}}$ the empirical operator-frequency distribution on the simplex $\Delta_{|\mathcal{F}|}$; the Euler–Lagrange equations on Δ give $2\mu\,\ddot{\bar{\mathfrak{p}}} - \lambda \nabla_{\bar{\mathfrak{p}}} H(\bar{\mathfrak{p}}) = 0$. Setting $V = -\log p(S)$, $T = -\log q(S \mid o)$ recovers Friston variational free energy as a special case.

Status: *Self-developed theory; operators & the relaxed action implemented.*

3.5 The base category

HCT's apparatuses are functors out of one base category Proc (objects: augmented states $(\tilde{\rho}, t)$; morphisms: operator-applications-with-memory-kernel) into specialized targets: $\mathcal{F}_{\text{quant}} : \mathsf{Proc} \to \mathsf{CPTP}$, $\mathcal{F}_{\text{geom}} : \mathsf{Proc} \to \mathsf{MetMeas}$, $\mathcal{F}_{\text{net}} : \mathsf{Proc} \to \mathsf{TensNet}$, $\mathcal{F}_{\text{prov}} : \mathsf{Proc} \to \mathsf{FreeMonoid}(\mathcal{F})$, $\mathcal{F}_{\text{var}} : \mathsf{Proc} \to \mathsf{Manifold}$.

Status: *Self-developed theory.*

4 The Physical Substrate (UMA / RSLS)

A stochastic–metriplectic field model used as a "crucible": a recovered law is mapped to a state and stress-tested under causal dynamics.

4.1 Causal flux and coupling

The Maxwell–Cattaneo constitutive law replaces parabolic diffusion with a hyperbolic (finite-signal-speed) flux,

$$\tau_J\,\partial_t J + J = -\,\mu\,\nabla\Phi,$$

where τ_J is the relaxation time and Φ the driving potential; the RSLS instantiation couples J to a memory field M through a singular barrier potential $V(M)$ and a Stage-6 self-consistency step.

Status: *Self-developed theory; integrators implemented & tested.*

4.2 Geometry: the Menger substrate

Admissible states are placed on a Menger sponge, whose Hausdorff dimension is

$$\dim_H = \frac{\log 20}{\log 3} \approx 2.7268.$$

Status: *Standard result (fractal dimension); used as substrate.*

4.3 Sensitivity: Lyapunov and predictability

The maximal Lyapunov exponent (Benettin renormalization) is

$$\lambda_{\max} = \lim_{t\to\infty}\frac{1}{t}\sum_k \log\frac{\|\delta_k\|}{\|\delta_0\|},$$

with a finite predictability horizon ("cone aperture") decreasing in $\lambda_{\max}$. The suite reports a frame-dragging contrast ($\lambda \approx +1.127$ chaotic vs. $\lambda \approx -0.044$ stable).

Status: *Standard method (Benettin); contrasts are proof-of-concept measurements.*

4.4 Open-system decoherence

The substrate's superposition evolves by the GKSL/Lindblad equation (as in T8), with von Neumann entropy

$$S(\rho) = -\,\mathrm{tr}(\rho\log\rho),$$

and a decoherence half-life tied in the bridge to the law's compression, $\gamma = \big(\max(0.2, \min(\rho_{\mathrm{cr}}, 20))\big)^{-1}$.

Status: *Standard (Lindblad, von Neumann); coupling to ρ_{cr} is proof-of-concept.*

4.5 Field-theoretic diagnostics

The substrate carries a complex Hamiltonian $H[\psi]$, an MSR response field, a Noether stress–energy tensor $T_{\mu\nu}$, and an Einstein consistency residual (deviation of the cognitive substrate from a linearized field equation); plus Koopman transfer operators, SRB histograms on the Lorenz system, the Kuramoto order parameter $re^{i\psi} = \frac{1}{N}\sum_j e^{i\theta_j}$, and distributionally-robust (Wasserstein-ball) value bounds.

Status: *Standard objects, instantiated; "Einstein residual" coupling is self-developed.*

5 Governance and Decision Calculus (JDICert / SoCPM)

5.1 The SoCPM decision rule

With context-sensitivity C_x, recommendation-autonomy A_r, human-impact H_p, mitigating-controls M_c, and value-alignment V, all in $[0, 1]$, define the gate

$$g \;=\; C_x A_r H_p - M_c(1 - V), \qquad \text{redirect/escalate if } g > T, \quad T = 0.25.$$

Status: *Self-developed rule; implemented & tested.*

5.2 Epistemic partition and Truth Horizon

Context is partitioned into Knowns K, Unknowns U, and the unknowable void Ω. The Truth Horizon is a bounded ratio of resolved to unresolved measure,

$$\Theta = \min\Bigl(\text{cap},\ \frac{m(K)}{m(U) + \varepsilon}\Bigr),$$

below a threshold of which the system must escalate.

Status: *Self-developed; implemented & tested.*

5.3 Confidence intervals and hallucination defeat

Belief deltas carry Hoeffding/PAC intervals: for bounded $X \in [a, b]$,

$$\Pr\left(|\bar{X} - \mu| \geq t\right) \leq 2\exp\Bigl(\frac{-2nt^2}{(b-a)^2}\Bigr).$$

Load-bearing claims are tested by Friston-style free energy $\mathcal{F} = \mathbb{E}_q[-\log p(o, s)] - H[q] = \text{surprise} - \text{entropy} + \mathrm{KL}(q\|p)$, together with a coherence score and a semantic-entropy ceiling; integrity is bound by HMAC-SHA256 attestation and a Merkle root over the certificate leaves.

Status: *Standard (Hoeffding, Friston, HMAC, Merkle); assembly is implemented & tested.*

5.4 Jurisprudence measures

Adversarial cross-examination accepts a rival generator as "reasonable doubt" when its MDL cost is within ε bits of the winner (*MDL parity*); precedent distance is Euclidean on the score vector (C_x, A_r, H_p, M_c, V); per-domain jurisdiction rescales the threshold T and the score weights. Provenance is a content-addressed DAG with a *minimal blame set* (smallest upstream commit set producing a challenged artifact).

Status: *Proof-of-concept; self-tested.*

5.5 Certification and the LLM accountability wrapper

A finding certificate composes the recovered law, a Daubert / FRE 702 reliability analysis (factors computed from the finding), a regulatory-compliance walk of the legal corpus, and a Merkle-root attestation with a replay seed. Crucially it states its own *legal status*: it is court-READY input, *not* a legal determination — admissibility is decided by a court, and the method's lack of peer review is disclosed in the analysis itself. The same discipline wraps a language model: given an LLM output, the wrapper audits its candor, *exactly verifies the checkable claims* (numeric sequences via collapse, arithmetic), marking each VERIFIED or REFUSED, runs the governance gate, and attests the result — but it never certifies a free-text claim as true. It certifies *accountability* (what is checkable, checked; the rest explicitly flagged unverifiable), not correctness. The model proposes; the engine disposes.

Status: *Implemented & tested; the certificate is an assurance artifact, not a legal instrument.*

6 Ambiguity and Origin (Infectatrum)

For a surface admitting a spectrum Σ of valid readings with probabilities p_i:

$$|\Sigma| \text{ (cardinality)}, \qquad H(\Sigma) = -\sum_i p_i \log_2 p_i \text{ (Shannon entropy).}$$

Origin attribution compares reading-distributions by Jensen–Shannon divergence

$$\mathrm{JSD}(P,Q) = H\big(\tfrac{P+Q}{2}\big) - \tfrac{1}{2}\big(H(P)+H(Q)\big) \in [0,1] \text{ bits,}$$

and an adversarial score discriminates engineered multiplicity (a deliberate flood) from natural noise. (Worked anchor: the IESVS/MARIA twin plates recover at $\mathrm{JSD} = \frac{1}{3}$.)

Status: *Standard (Shannon, JSD); implemented & tested on the Caramuel corpus.*

7 Derived Measures (the bridge modules)

7.1 Mathematical beauty (aesthetics)

$$\begin{aligned}
\text{Elegance} \quad & \mathcal{E} = 1 - \rho^{-1} \in [0,1] \quad \text{(MDL saturation)}, && (10)\\
\text{Symmetry} \quad & \mathcal{S} = \max\big(\mathrm{corr}(x, \overleftarrow{x}),\ \max_{\ell} \mathrm{ac}(x,\ell),\ 2 - \mathrm{FD}_H(x)\big), && (11)\\
\text{Resonance} \quad & \mathcal{R} = \tfrac{1}{2}\Big(\tfrac{1}{1+e^{2\lambda_{\max}}} + (1 - \mathrm{SF}(x))\Big), && (12)\\
\text{Beauty} \quad & \mathcal{B} = (\mathcal{E}\cdot\mathcal{S}\cdot\mathcal{R})^{1/3}, && (13)
\end{aligned}$$

where FD_H is the Higuchi fractal dimension (slope of $\log L(k)$ vs. $\log(1/k)$), ac is lag autocorrelation, and the spectral flatness is the ratio of geometric to arithmetic mean of the power spectrum, $\mathrm{SF}(x) = \big(\prod_j P_j\big)^{1/N} / \big(\frac{1}{N}\sum_j P_j\big)$. The geometric mean makes $\mathcal{B}$ unforgiving: one low factor sinks it.

Status: *Self-developed composite of standard measures; implemented & tested.*

7.2 Value and harmony (ontological)

$$V = \tfrac{1}{3}\big(\mathbf{1}[\text{verified}] + (1-\rho^{-1}) + (1 - \tfrac{|\text{resid}|}{n})\big), \qquad \mathcal{H}(v) = \max\Big(0,\ 1 - \min\big(1, \tfrac{\sigma(v)}{|\mu(v)|}\big)\Big),$$

value as the mean of verification, compression saturation, and fidelity; harmony as one minus the normalized dispersion of a set of measures.

Status: *Self-developed; implemented & tested.*

7.3 Language (offline, no model)

$$
\begin{aligned}
\text{PPMI:}\quad & \mathrm{ppmi}(a,b)=\max\Big(0,\ \log\tfrac{P(a,b)}{P(a)P(b)}\Big),\quad \cos(u,v)=\tfrac{\langle u,v\rangle}{\|u\|\,\|v\|}, && (14)\\
n\text{-gram (add-}k\text{):}\quad & P(w\mid c)=\tfrac{\#(c,w)+k}{\#(c)+k\,|V|},\quad \text{perplexity}=2^{H}, && (15)\\
\text{Yule's } K:\quad & K=10^4\,\tfrac{\sum_i i^2V_i-N}{N^2},\qquad \text{Simpson } D=\tfrac{\sum_w c_w(c_w-1)}{N(N-1)}, && (16)\\
\text{Flesch RE:}\quad & 206.835-1.015\,\tfrac{\#w}{\#s}-84.6\,\tfrac{\#\text{syll}}{\#w}, && (17)\\
\text{Burrows } \Delta:\quad & \Delta(u,c)=\tfrac{1}{|\mathcal{W}|}\sum_{f\in\mathcal{W}}\big|z_f(u)-z_f(c)\big|, && (18)
\end{aligned}
$$

with function-word z-scores z_f, plus Gunning fog and Coleman–Liau readability. Distributional similarity, injection/anomaly detection (leave-one-out surprisal), and Burrows authorship are all computed offline from local text.

Status: *Standard measures; implemented & tested (17/17 gates), proof-of-concept on real prose.*

8 Combinatorics of the Caramuel Atlas (Primus)

The *Primus Calamus* labyrinths are read as constrained permutation spaces. A proteus verse permutes a fixed multiset of words and admits only those orderings that scan as a dactylic hexameter; the metrical constraint

$$\text{line} \models \text{hexameter} \iff \text{the syllable-quantity string matches } (\mathrm{D}|\mathrm{S})^4\,\mathrm{D}\,\mathrm{S},$$

(D = dactyl $-\cup\cup$, S = spondee $--$) filters the orbit. The atlas anchor is a combinatorial surface space of order $\sim 2.8\times10^{17}$ collapsing to shared generators; Chiron recovers cross-plate twins (identical generator, divergent surface).

Status: *Standard combinatorics + classical prosody; scansion implemented & tested.*

9 The Projection–Innovation Hierarchy (PIH)

A single real action on a doubled state $(q,\hat q)$ (Martin–Siggia–Rose), $\mathcal{S}_{JD}[q,\hat q]$, such that

- its first variation gives the deterministic dynamics;
- its second variation gives the noise covariance;
- its non-symmetric part gives entropy production;
- its parameter Hessian gives the Fisher information metric;
- its exponential generates the Onsager–Machlup path measure $p[q]\propto\exp(-\mathcal{S}_{JD})$.

Iterating the MSR doubling yields a hierarchy of fields $(q,\eta_1,\eta_2,\dots)$, well-posed at each finite level, with explicit limits to the Kalman filter, Schwinger–Keldysh field theory, classical mechanics, and stochastic thermodynamics.

Status: *Self-developed theory (companion document); not implemented in the engine.*

10 External benchmarks (one architecture, six tasks)

The recovery schema is exercised on six independent tasks — four recovery domains plus two head-to-head comparisons against established methods — each beating or matching a standard baseline and refusing rather than guess where refusal applies.

task	result	baseline	refusal
integer sequences	8/8 recovered, 8/8 held-out exact	gzip / bz2 / lzma	0 false-verify
proverb semantics	2/2 invariants, controls refused	bag-of-words	refuses controls
protocol / FSM	2/2 machines, 240/240 held-out	first-order Markov	abstains on noise
governance	4/4 scenarios, citations attached	domain-blind search	abstains off-scope
symbolic regression	6/7 exact	polynomial regression (3/7)	refuses control
authorship	Burrows Δ above chance, deterministic	content baseline	—

Protocol recovery is a fresh instantiation of the schema: deterministic finite-state machines are the hypothesis class, MDL prefers the minimal transition table consistent with the observed traces, held-out traces are predicted at exact equality, and contradictory (non-deterministic) observations are *refused* rather than forced into a wrong machine. The two comparisons make the contrast with standard practice explicit: against polynomial regression the engine recovers non-polynomial laws that least-squares gets confidently wrong and *abstains* on a structureless control rather than emit a confident answer; the authorship task validates an established stylometric method (Burrows's Δ) on held-out passages. A general-purpose compressor shrinks the bytes it is shown but cannot predict an unseen term; the recovered law can — the architecture's payoff is prediction, not merely size.

Status: *Implemented & tested (suite verdict* PASS*, 6/6 tasks).*

11 The unifying object

Every section above instantiates one schema. Let a surface s live in a space with a description-length functional L, a hypothesis family $\mathcal{H}$, and a recognition equivalence $\sim$. The portfolio's core operation is

$$\boxed{M^\star = \arg\min_{M \in \mathcal{H}} L(s, M), \quad \text{accept } M^\star \iff M^\star \text{ predicts held-out } s \text{ exactly}, \quad \text{else } \textsc{incompressible}.}$$

Integers and ciphers (§1), meaning (§2), identity under transformation (§3), dynamical survival (§4), admissible action (§5), and ambiguity/origin (§6) are the same arg min-plus- refusal read in different spaces. The defining property is not maximal recall but the discipline of declining to certify what cannot be exactly verified. The same gate admits a probabilistic model safely: when a language model is used to *propose* candidate structures, the exact verifier still *disposes*, so the zero-false-positive floor is preserved even with a model in the loop. This contract is realized as a single interface (`epistemic.py`): Surface → Hypothesis Space → Constraint Space → Verification → Certificate, with the integer engine, the semantic calculus, the governance layer, and an energy layer as four instances of it. The energy layer (`semic_energy.py`) sits *above* exact collapse, not in place of it: exact collapse certifies or refuses, and only on refusal does a Gibbs measure over the generator space return an explicitly *uncertified* approximation — exact symbolic recovery with approximate exploration around it.

Status: *The schema is the genuine, load-bearing contribution.*

11.1 The whole spine in one file

The code that realizes this schema across sixty-two modules is also distributed as a single file. `chiron_monolith.py` embeds the byte-identical source of every Chiron module and wraps them in an import hook (`sys.meta_path`) so that each inter-module import resolves to the embedded copy; any module then runs from the one file (`python3 chiron_monolith.py ⟨module⟩ ...`). The fold is *lossless*—each embedding is asserted byte-identical to its origin at build time—and adds no logic: the gates that pass for the standalone scripts pass unchanged when run through the fold (the core-engine battery and every selftest-bearing module verify identically). The single-file property long held by the integer engine `chiron.py` is thereby extended to the entire portfolio.

Status: *Implemented & tested: a lossless single-file fold of the spine, not new mathematics.*

Notation

$L(s, M)$	two-part description length of surface s under generator M
$M^\star,\ I(F)$	recovered minimal generator / semantic invariant
$\rho(s),\ \mathcal{E}$	compression ratio; MDL saturation $= 1 - \rho^{-1}$
$\mathrm{Res}(O, S) = \lvert O \triangle S \rvert$	structural residual (symmetric-difference size)
$p_T,\ F(T)$	Gibbs measure at temperature T; free energy
$\kappa(x, y)$	Ollivier–Ricci curvature; W_1 the 1-Wasserstein distance
$\pi, \rho,\ \mathfrak{p}$	projection, recovery, provenance word
$\xi \in \mathbb{R}^6$	Mystery Vector (six species of the unknown)
$g,\ T{=}0.25$	SoCPM gate and threshold
$\Theta,\ K/U/\Omega$	Truth Horizon; Knowns / Unknowns / void
JSD, H	Jensen–Shannon divergence; Shannon entropy
$S(\rho) = -\operatorname{tr}(\rho \log \rho)$	von Neumann entropy
$\lambda_{\max}$	maximal Lyapunov exponent
$\mathcal{B} = (\mathcal{E}\mathcal{S}\mathcal{R})^{1/3}$	aesthetic (beauty) score

References (standard results used)

1. Solomonoff, R. (1964); Kolmogorov, A. (1965) — algorithmic complexity.
2. Rissanen, J. (1978) — minimum description length.
3. Singleton, R. C. (1964) — maximum-distance codes (the rate bound, T7).
4. Lindblad, G. (1976); Gorini–Kossakowski–Sudarshan (1976) — CPTP semigroups (T8).
5. von Neumann, J. — density matrices and entropy.
6. Ollivier, Y. (2009) — Ricci curvature of Markov chains; Wasserstein W_1.
7. Friston, K. (2010) — the free-energy principle / active inference.
8. Benettin et al. (1980) — Lyapunov exponent computation.
9. Cattaneo, C. (1948) — hyperbolic (finite-speed) heat conduction.
10. Almheiri–Dong–Harlow (2015); Pastawski et al. (2015) — holographic QEC.

11. Martin–Siggia–Rose (1973); Onsager–Machlup (1953) — stochastic field actions (PIH).
12. Mosteller & Wallace (1964); Burrows (2002) — stylometry / authorship.
13. Austin, J. L. (1962) — performative speech acts (Class VIII).

Author: Jacob Iannotti. The exact-recovery core (§1) and the measures marked *implemented & tested* are reproducible and covered by self-tests; items marked *self-developed theory* are constructive and not peer-reviewed; items marked *standard result* are used as-is. This document states each at its true resolution.